ModelIt!

Student Activity Pack

Can You Predict the Weather?

Patterns in Weather and Climate

Grade 3 | NGSS 3-ESS2-1 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the weather like today? How do you know without looking outside?

2 If it is July, can you guess if it will be hot or cold? How do you know?

3 Draw your favorite kind of weather and explain why you think it happens.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Temperature | Cloud Cover | Chance of Rain

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Hot Summer Day

Temperature set to high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cold Winter Day

Temperature set to low

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cool Spring Day

Temperature set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Speed — How fast the air is moving — wind can bring in new weather from other places

Humidity — How much water vapor is in the air — more humidity means more chance of rain

Season — What time of year it is — seasons create the biggest weather patterns

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can we figure out what the weather will be like tomorrow by looking at patterns from the past?
Use evidence from your simulation to support your answers.

1 How are weather and climate different, and why does knowing the difference matter?

2 How did your model help you understand why meteorologists can predict the weather?

8 STEM Challenge: Build a Class Weather Station

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YOUR ENGINEERING MISSION

Design and build a simple weather station that tracks temperature and cloud cover to make weather predictions for your school.

THE SCENARIO

Your school principal wants students to predict the weather for morning announcements each day. Your team must design a weather tracking station that collects data and uses patterns to make forecasts. Use what you learned about weather patterns to build it!

Engineering Design Thinking

1. WHERE IS THE BEST PLACE TO PUT YOUR WEATHER STATION SO YOU GET ACCURATE READINGS?
2. HOW WILL YOU MEASURE TEMPERATURE — WITH A REAL THERMOMETER OR A WEATHER APP?
3. WHAT CHART OR GRAPH WILL YOU USE TO TRACK DATA OVER TIME?
4. HOW WILL YOU SHARE YOUR WEATHER PREDICTIONS WITH THE SCHOOL?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists study weather patterns and use science to predict storms, temperature, and precipitation. They earn $50,000-$100,000/year.